

Design of a 2D and 3D Situation Display Platform Based on WebGL and Modern Web Technology Stack

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Abstract—This paper aims to discuss and introduce a comprehensive situation display platform built on WebGL and modern Web technology stack and its implementation process. Based on the unified information service standard specification, the platform completes the integration and organization of various types of data with multi-source, heterogeneous, and decentralized storage. It provides users with rich geographic information services and an immersive data visualization experience. This paper describes the system architecture, technical features, and the realization of functional modules of the platform in detail, which reflects the advantages of using geographic information efficiently and improving user experience.

Keywords—situation display platform; WebGL; Web technology stack.

I. INTRODUCTION

With the increasingly severe international situation and higher technological level, the application of computer simulation technology in the military field is becoming more widespread [1]. The situation display system mainly plots the information required for combat command on the screen in a real-time and intuitive form, which can reflect the dynamic changes of the battlefield in a timely manner and provide a visual comprehensive situation display interface for commanders or combat personnel to assist command decision-making and enhance battlefield awareness. It is an important component of combat command and decision-making [2].

Therefore, this article proposes a construction scheme for a comprehensive situation display system based on WebGL and modern web technology stack. The comprehensive situation display tool is a lightweight 2D and 3D geographic information platform that is cross platform and cross browser. It integrates visualization libraries and graphics libraries, supports hardware accelerated visualization rendering technology, and integrates 2D and 3D display functions into a situation visualization system. At the same time, the linkage mechanism of the 2D and 3D display parts across platforms has been studied [3]. The comprehensive situation display tool loads models and corresponding scene data based on the experimental plan to construct a three-dimensional visualization environment. During the experimental operation, the comprehensive situation display tool drives scene elements to complete the visualization display of the experimental system. After the experiment is completed, the comprehensive situation display tool uses

playback data to reproduce the experimental process under the drive of experimental data collection, storage, and playback software.

II. PLATFORM ARCHITECTURE DESIGN

A. Service-Oriented Architecture Design

The technical architecture design of this system is based on the actual application requirements of the two-dimensional and three-dimensional situation display software, adopts the service-oriented architecture (SOA) concept, combines the business requirements of resource management, software development technical specifications, and hardware deployment environment conditions to carry out hierarchical design, including infrastructure layer, data storage layer, service layer, and an application interaction layer, as shown in Fig. 1.

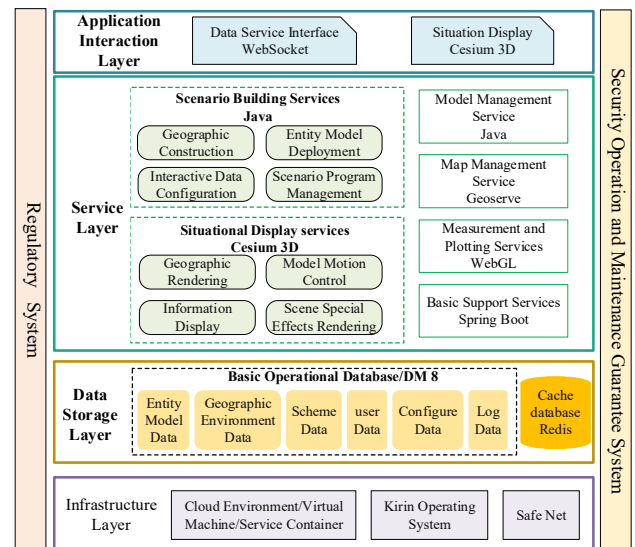


Figure 1. Software Architecture

The infrastructure layer provides the hardware and software environment necessary for system deployment and operation. The data storage layer manages the storage of resource data and essential operational data. The service layer utilizes a micro services architecture to deliver unified foundational and application services, ensuring high cohesion and low coupling for better system modularity. The application interaction layer

offers user portals and interaction interfaces for external systems, facilitating seamless communication and system integration.

B. Technical Architecture

As a universal geographic environment simulation platform, GIS (Geographic Information Services) can easily simulate situational environments. The basic functional requirements of the comprehensive situation display system can be achieved by conducting secondary development of the GIS system, adding situation target display function, and establishing relevant databases and network communication components [4]. The GIS platform can be categorized from two perspectives: functionality and logic.

From a functional perspective, the platform consists of three layers: the user layer, the business layer, and the data layer. The business layer is further divided into three components: the JavaScript engine, web services, and map services. From a logical perspective, the platform is divided into two parts: the front end and the back end. The front end comprises the client web map and the JavaScript engine, responsible for user interactions and forming the web map application. The back end includes web services, the map server, and the database, managing data and processing requests without direct user interaction.

The interaction process between layers is as follows: When users perform operations or request data in the web map application (browser), the JavaScript engine sends a data request to the web server in the business layer. The web server, through the GIS proxy server, parses the request into commands that the map server can process and sends corresponding database queries to the data layer. The data layer executes the queries, returning spatial or statistical data. The GIS proxy server reorganizes the data, reducing the client's processing load, and passes it back to the web server, which delivers the data to the client, completing the data loading process. The technical architecture of the GIS platform is illustrated in Fig. 2.

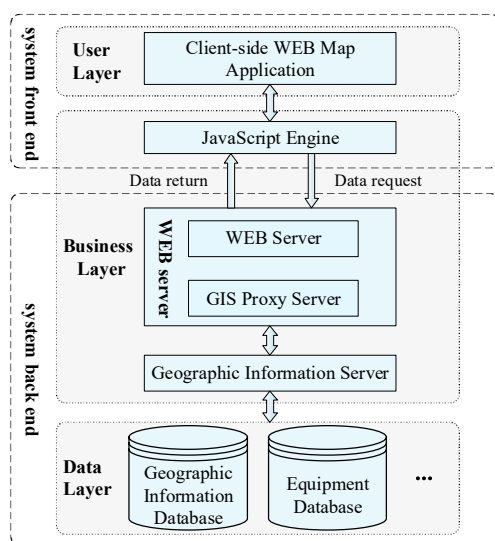


Figure 2. System Technical Architecture

The platform separates the computing and request distribution functions of the server, which makes the data request processing of the client more efficient, improves the user experience through the use of Web technology, and enhances the ease of use of the system. The platform also designed a data storage mechanism based on distributed technology in the data layer, and deployed different types of data on different hosts, thus reducing the burden of the database.

C. Front-end Architecture

Fig. 3 shows the front-end hierarchy of the GIS platform. Each layer provides an implementation basis for its upper layer, and each layer is an abstract implementation of its lower layer. In this way, the front-end Web map application of the system is finally realized.

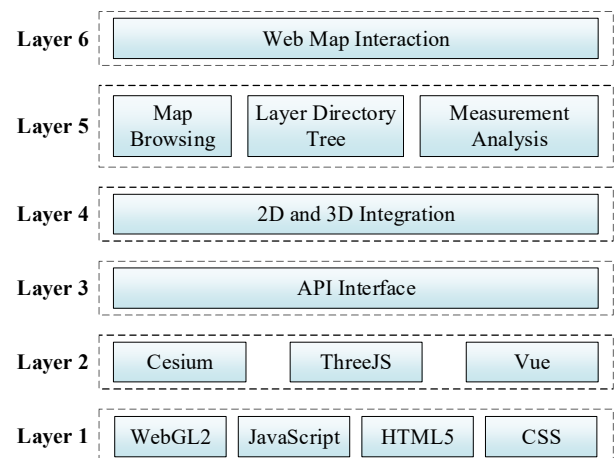


Figure 3. Front-end Architecture

The first layer includes foundational technologies such as WebGL, JavaScript, HTML5, and CSS. These technologies form the core of the web mapping application. As a 3D drawing standard, WebGL is implemented based on a standardized, cross platform, and unified OpenGL interface, which can improve the speed of hardware 3D rendering for Canvas in HTML5. This allows developers to use the help of underlying hardware graphics cards to render 3D geographic scenes in the browser, and WebGL is implemented through web page scripts without loading any plugins [5].

The second layer is Cesium and ThreeJS class libraries, as well as the Vue front-end framework. Cesium is an open-source JavaScript library used for creating map services in web browsers. Cesium can smoothly support the latest HTML5 browsers without the need to install additional extension plugins, which is suitable for the development needs of lightweight WebGIS users and can effectively improve user experience [6]. Vue is a JavaScript front-end framework, which makes it easier to develop Web single page SPAs (Single Page Applications) applications, and also helps to achieve front-end modular development in this system [7].

The third layer is the API interface layer. The interface layer is the sub-packaging of the interfaces of Cesium and OpenLayers, which integrates the functions of two-dimensional maps and three-dimensional maps into the same

interface. When the API encapsulated by the interface layer is called in the front-end module, the internal API will automatically synchronize the map operation or data loading to the two-dimensional and three-dimensional maps.

The fourth layer includes integrated 2D and 3D functional modules built using Cesium and ThreeJS. These modules ensure operational linkage and data synchronization between 2D and 3D maps, providing a framework for further functional module development and enabling core system functionalities like 2D/3D synchronization.

The fifth layer is the realization of the system display function module. The front-end functional modules in the system include a map browsing module, layer management module, measurement analysis module, etc. The map browsing module realizes the loading and switching of map data and the searching and positioning of maps according to the grid number. The measurement and analysis module realizes the measurement of distance, area, and azimuth in the map, as well as the spatial analysis functions such as slope, slope direction, profile, and view-through.

The sixth layer is the Web map interaction layer, which builds the basic framework of the Web page through HTML tags and adds styles to the page tags through CSS to make the page more beautiful. It receives geographic data from the server and visualizes it on the web page. This layer listens for user interactions with the map and updates geographic data in real-time through JavaScript, ensuring a dynamic and responsive user experience.

D. System and Network Deployment Structure

The system is a geographic information platform based on B/S architecture, which is divided into server side and browser side. The system deployment structure is shown in Fig. 4.

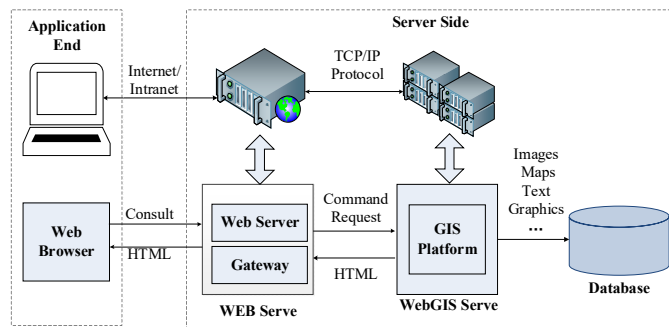


Figure 4. System Deployment Structure

Web Browser: Users can access various geographic information data published by the server through a web browser.

Web Server: This server sets up the geographic information proxy mechanism and protocol, providing data access interfaces. It is crucial for the publication of geographic information.

Web GIS Server: This server operates the GIS database based on user requests, providing GIS to users and enabling dynamic interaction between the client and the server.

The GIS platform adopts a B/S (Browser/Server) architecture with a centralized deployment model. It supports

deploying the platform on standard servers, portable servers, and standalone environments, as shown in Fig. 5.

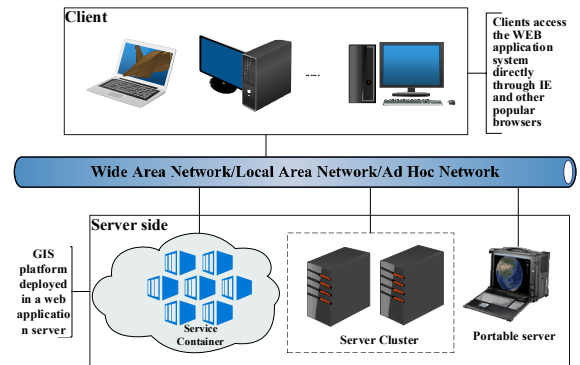


Figure 5. System Deployment Structure

Deployment on Standard Servers: The GIS platform is deployed on standard servers, providing on-demand web application support to users in the task area via a local area network (LAN).

Deployment on Portable Servers: The GIS platform is deployed on portable servers, allowing for ad-hoc networking to provide front-end web application services to mobile troops.

Deployment in Standalone Environments: The GIS platform is deployed in a standalone environment, requiring no network connectivity, and enabling simple functional applications on the web interface.

III. FUNCTION DESIGN

A. Functional Components

The 2D and 3D situation display software is composed of six functional components, as shown below.

Scene Construction Module: It mainly includes geographic environment, natural environment and environment marking. In the construction of geographic environment, it is necessary to select the loading area required for this test, so that the subsequent loading of the perspective directly display the scope of the test.

Situation Display Module: It mainly includes real-time data drive, dynamic posture display, display control, mapping analysis. It carries out the display of combat mission scenes and various physical facility elements, and drives the visualization display of the overall posture process based on data to vividly show the battlefield confrontation environment.

Model Management Module: It mainly includes management of basic geographic information data and entity model libraries.

Map Management Module: It mainly includes basic geographic information data management functions, divided into offline map data management, layer management, map base service management sub-module.

Measurement and Mapping Module: It is divided into two sub-modules: Measurement Analysis Tool and Scene Mapping Tool.

area, and inserts the relationship between the scenario and the geo-environmental resource data. After successful operation, the database operation result will be returned, and finally feedback the interface operation prompts to the scenario constructor.

When the scenario builder carries out the operation of exporting scenarios, the data storage area is accessed to obtain and return the scenario data, and the scenario builder's scenario data file is returned through the data conversion of the scenario program management.

C. Situational Function Design

The posture display module displays the combat mission scenario and various physical facilities, and through real-time data-driven visualization of the situation process, vividly presents the battlefield confrontation environment. It loads the pre-set combat environment and, based on the real-time data pushed by the data interaction module, demonstrates the real-time progress of the enemy friendly situation and combat actions according to the combat plan, as shown in Fig. 8.

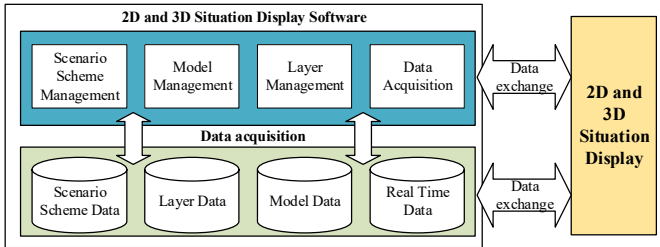


Figure 8. Data Flow Diagram for Situational Display

D. Map Management Function Design

The layer management class diagram is shown in Fig. 9. The map management function module mainly designs the entity class TbMapLayer, control class TbMapLayerController, layer management business processing class TbMapLayerService, engineering and layer relationship management business processing class TbProjectLayerService, layer management data access operation interface class TbMapLayerMapper, interfaceMPJBaseService, interface MPJBaseMapper, and service implementation classMPJServiceImpl.

The entity class TbMapLayer is responsible for recording the name and characteristics of scheme management; TbMapLayerController is a layer management access control class. It provides abstract methods to implement data interface modification, addition, deletion, etc. by obtaining data details interface, and obtains or modifies project data by calling the method of TbMapLayerService. It relies on TbProjectLayerService to ensure that the request for project and layer relationship management business can be processed correctly. TbMapLayerServiceimplements its business logic by relying on TbMapLayer, TbMapLayerMapper and MPJServiceImpl. It uses TbMapLayer as the data model, accesses data through TbMapLayerMapper, and may call the functions provided by MPJServiceImpl, thus realizing effective management and operation of map layer data. This design pattern enhances the collaboration between various components and improves the flexibility and maintainability of the system.

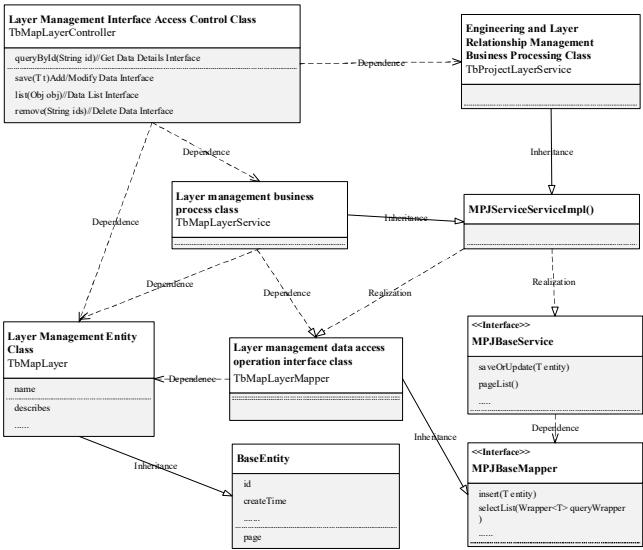


Figure 9. Layer Management Class Diagram

IV. SOFTWARE EFFECTS

This section presents some screenshots of the 2D and 3D situational display software. First, it showcases a schematic diagram of the scene loading in the situational display module and a schematic diagram of the 3D scene map tool control, as shown in Fig. 10.

The operators can click on the Entity Model Library to display each entity model in the display area on the right side of the system, as shown in Fig. 11.



Figure 10. Schematic Diagram of the 3D Scene Map Tool Control

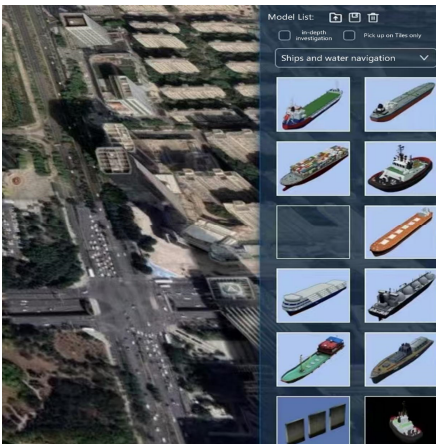


Figure 11. Schematic Diagram of the Entity Model Library

Operators can click the base-map switch button on the left side, which will display the supported base-maps in the system on the left side of the interface, as shown in Fig. 12. There are various base-maps available for operators to choose from, facilitating a more intuitive observation of the map interface under different conditions.



Figure 12. Schematic Diagram of the Basemap Switching Interface

V. CONCLUSION

This paper introduces the design and implementation of a two-dimensional and three-dimensional situation display platform based on WebGL and modern Web technology stack, aiming at providing a cross-platform real-time response and multi-functional integration situation display tool to meet the needs of complex geographic information systems.

The design of the platform includes a unified service architecture combined with service architecture and modular design, which makes the system have the advantages of high cohesion and low coupling, and ensures the flexibility and scalability of the platform. The core functions of the platform

cover the interactive scene construction of 2D and 3D maps, data management, dynamic situation display and measurement modules, etc. It provides powerful functional support for users and can meet the application requirements of various fields. Through the separation of front-end and back-end architecture platform, the performance and user experience are optimized, which can realize fast geographic information rendering in browser and efficient data interaction with map server through Web services. The system supports a variety of data types and layer management, and users can display the content according to their needs, which greatly improves the operability and real-time performance of the system.

Generally speaking, this platform has significant advantages in improving the utilization efficiency of geographic information, enhancing user interaction experience and realizing efficient situation and decision support. In the future, with the continuous progress of technology and demand, the platform can further expand its functions to support applications in more fields and provide solid technical support for various scenarios such as urban planning, monitoring and military command.

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